Digital Communication System

Unit 5 – New Generation Mobile Services

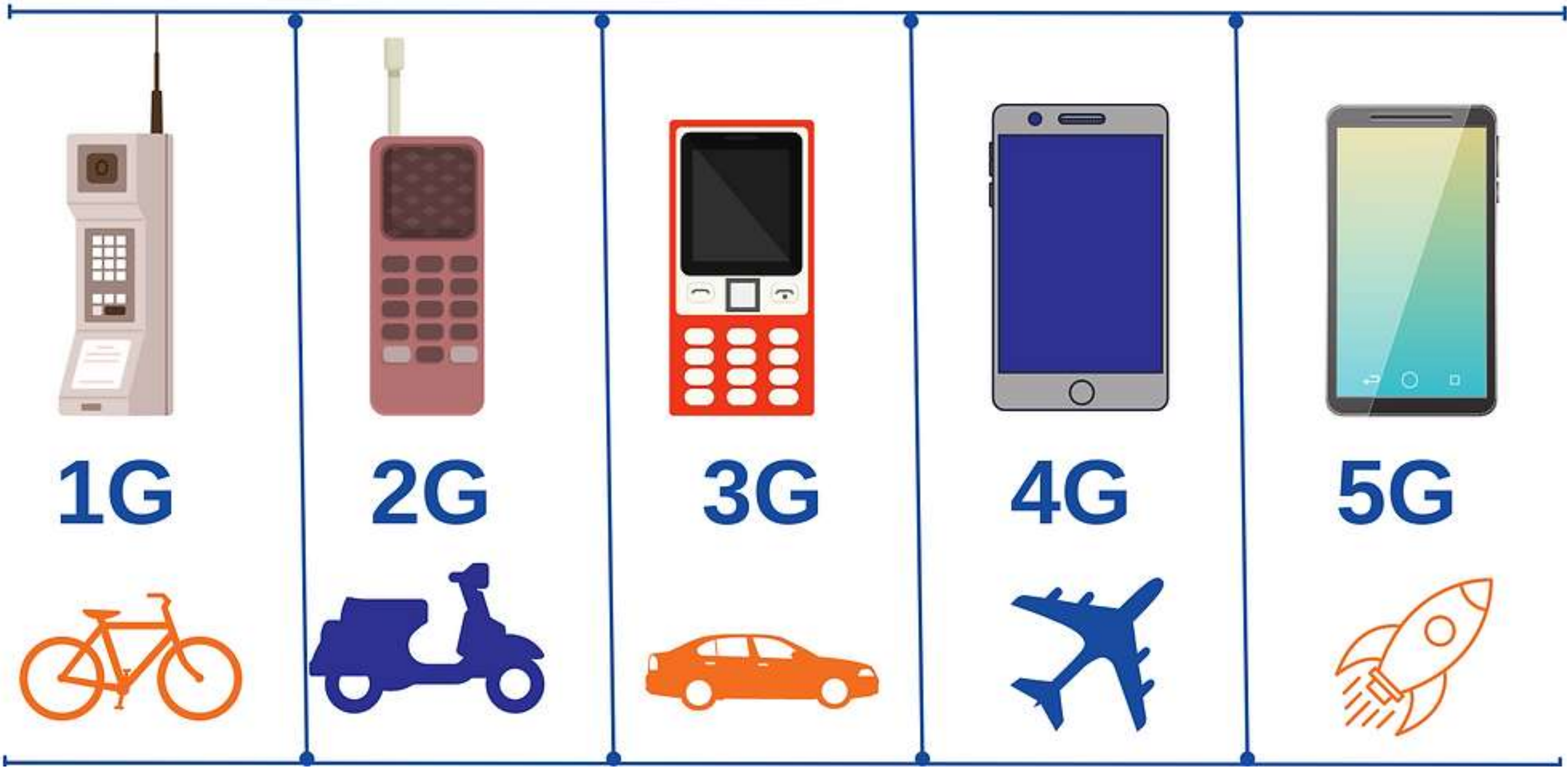
5.1 Introduction to 4G technology

5.2 Introduction to 5G technology

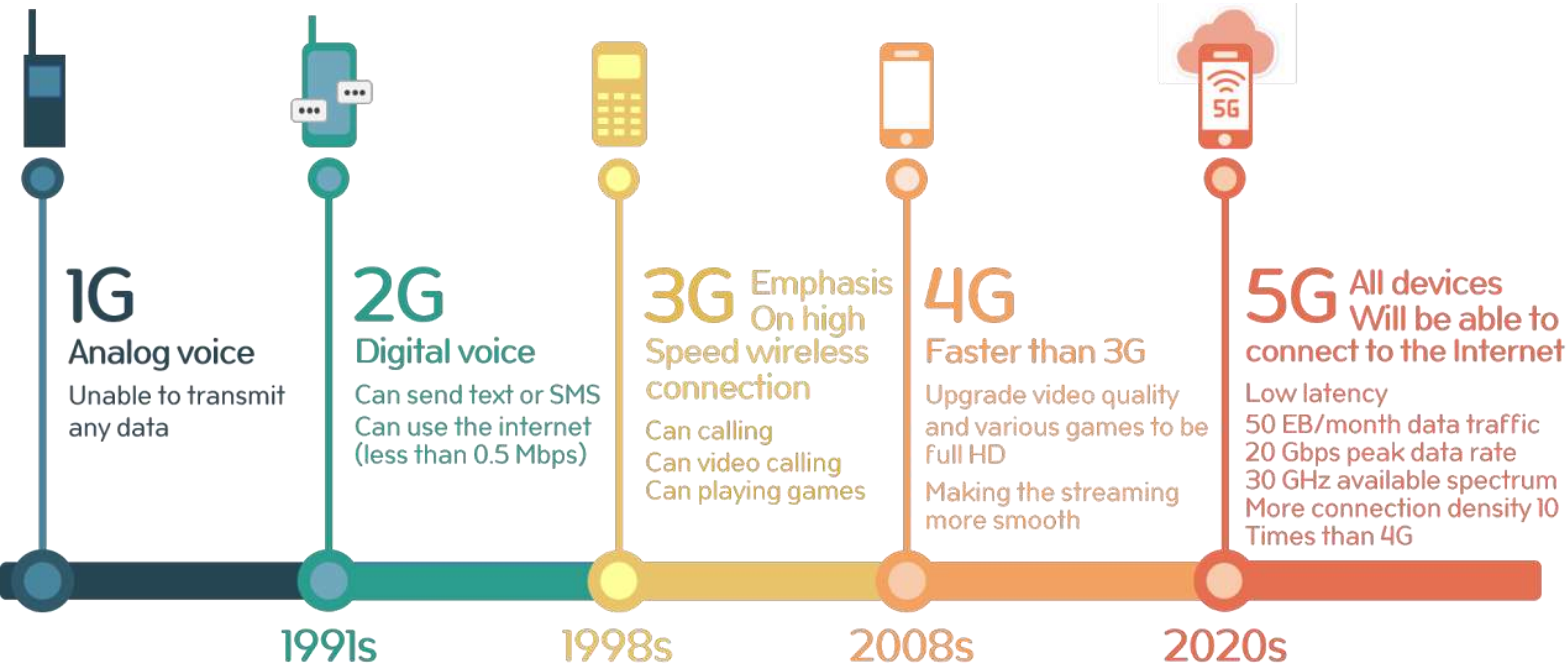
5.3 Introduction of Internet of Things

5.4 IoT / M2M Applications

Generations of Cellular Communication Technology



Generations of Cellular Communication Technology



1G or AMPS(Advanced Mobile Phone System)

•Characteristics:

•Used Analog Signals.

•Supports only voice calls.

•Commercially introduced in U.S. in 1980s.

•It used **FDMA** (Frequency Division Multiple Access) technology.

•Limitations:

•No support for SMS or Data Service.

•Limited battery life.

•Poor voice quality.

•No security.

•Speed supported by it was 2.4Kbps. (Very slow)

2G or GSM(Global System for Mobile Communication)

•Characteristics:

•Used Digital Signals.

•Supports voice calls and data transmission.

•Noise was less due to digital transmission.

•Used CDMA(Code Division Multiple Access) and TDMA(Time Division Multiple Access) technology.

•Advantages over 1G:

•Better quality than 1G due to noise immunity.

•Better bandwidth utilization.

•Speed was upto 64Kbps.

•SMS and multimedia was possible.

2.5G or GPRS(General Packet Radio Service)

- **Characteristics:**
- Add-on to 2G.
- Enabled web browsing, email, fast upload download speed.
- 2.5G was based on packet switching technology whereas 2G was based on circuit switching technology. It reduced the call and data cost drastically.

3G

- **Characteristics:**
- Supports much higher data transmission rates.
- Increased bandwidth.
- Uses Packet switching techniques like 2.5G.
- Speed was upto 144Kbps.

- **Advantages over 2.5G:**
- High speed data and voice transmission.
- 3D gaming was possible.
- Mobile Apps were possible.
- Fast and easy transmission of multimedia files.

4G

•Characteristics:

•10 times faster data rates than 3G.

•Speed is 100Mbps or beyond.

•Competing 4G standards are based on Mobile WiMax and LTE(Long Term Evolution) Technology.

•**OFDM** (Orthogonal Frequency Division Multiplexing) and **MIMO** (Multiple Input Multiple Output) technology are used.

•Packet and message switching is used.

•Carrier frequency is 15MHz.

•Advantages over 3G:

•Very high speed data and voice transmission.

•Mobile Apps and Web based apps are accessible.

•Makes Mobile broadband possible.

5G

•Characteristics:

•Still under introduction phase in India.

•Ultra high data capability.

•Speed upto 1Gbps. (10 times faster than 4G)

•Based on packet and message switching.

•Advantages over 4G:

•Simultaneous access to various wireless technologies. IoT and exponential data growth is supported by 5G.

•Fast response time (approx. 1millisecond) compared to 4G(60 - 98millisecond) and faster download speeds.

•Limitations of 5G:

•Doesn’t have coverage everywhere.

•Requires advanced hardware device.

•Mobile battery drains faster compared to 4G as mobile network shifts from 4G to 5G and 5G to 4G due to less coverage of 5G.

4G Introduction

- **4G -** 4G, the fourth-generation of wireless service, is an enhancement from 3G and is presently the most extensive, widespread, expeditious and high-speed wireless service.

- For instance Sprint employs a technology called **WiMax f**or its 4G services, whereas Verizon Wireless employs Long Term Evolution, or **LTE (Long Term Evolution).**

- On an average, 4G wireless technology is expected to provide data rates from four to ten times higher than conventional 3G networks.

4G

- **4G -** The vital characteristics of the 4G networks include accessing information with a flawless connection anytime, anywhere with a wide range of services, receiving greater amounts of information, pictures, data, video, and so on.

- 4G focuses on ensuring a flawless service, have larger bandwidth, higher data rates, and smoother and faster handoff across a wide range of wireless networks and systems.

- **Fast Data Rate than 3G (10 times faster)**

- **Makes Mobile Broadband possible**

Characteristics of 4G

- Wider and extensive mobile coverage region.
- Larger bandwidth - higher data rates(100Mbps or more)
- Seamless switching across heterogeneous networks.
- Speed is very high Global roaming and inter-working among various other access technologies.
- Suitable for HD video calling, 3D online gaming, live streaming, etc.
- Packet Switching. (High Speed at Low cost)
- Carrier frequency is 15 MHz.

Characteristics of 4G

- 4G capabilities for transmitting and receiving the signal are based on **OFDM** (Orthogonal Frequency Division Multiplexing) and **MIMO** (Multiple Input Multiple Output) technology.

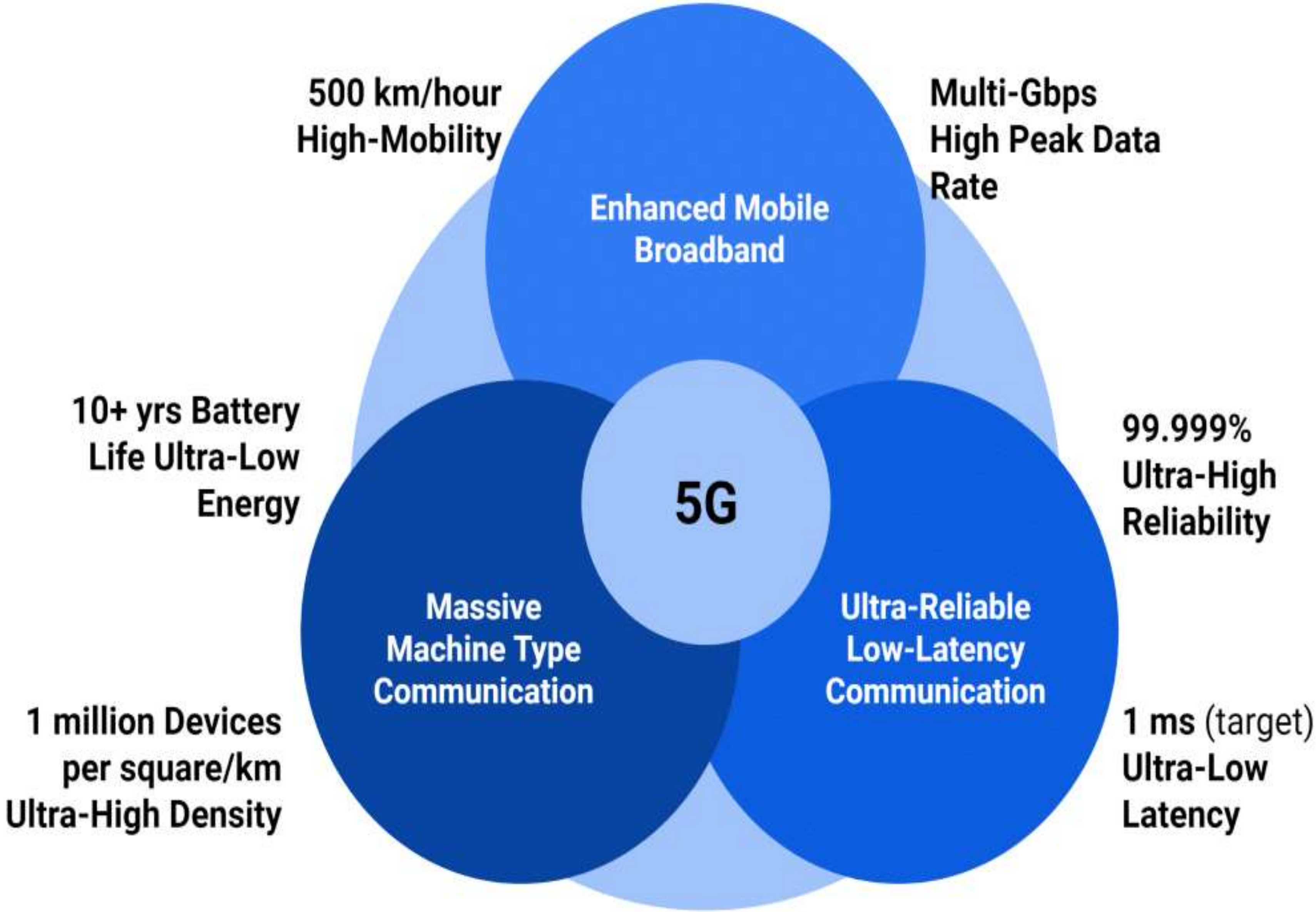
- MIMO technology reduces network congestion as compared to 3G and hence services can be served to more users without network congestion.

Advantages of 4G

- Accessibility of mobile and web apps, Mobile TV, etc.
- Cost of Internet Services has fallen down compared to 3G
- Cloud Services are possible.

Disadvantages of 4G

- Depends on the signal strength
- It needs more battery consumption


500 km/hour
High-Mobility
Multi-Gbps
High Peak Data
Rate
Enhanced Mobile
Broadband
10+ yrs Battery
Life Ultra-Low
Energy
5G
99.999%
Ultra-High
Reliability
Massive
Machine Type
Communication
Ultra-Reliable
Low-Latency
Communication
1 million Devices
per square/km
Ultra-High Density
1 ms (target)
Ultra-Low
Latency

5G Introduction

- 5G is the fifth generation of wireless cellular technology, offering higher upload and download speeds, more consistent connections, and improved capacity than previous networks.

- 5G is much faster and more reliable than the currently popular 4G networks and has the potential to transform the way we use the internet to access applications, social networks, and information.

- For example, technologies like self-driving cars, advanced gaming applications, and live streaming media that require very reliable, high-speed data connections are set to benefit greatly from 5G connectivity.

Why is 5G important?

- The demand for internet access, combined with the emergence of new technologies such as artificial intelligence, the Internet of Things (IoT), and automation, is driving a massive increase in the amount of data created.

- Data creation is growing exponentially, with volumes set to increase by several hundred zettabytes over the coming decade.

- The current mobile infrastructure was not designed for such a high information load and requires upgrading.

Why is 5G important?

- At the same time, with its high speed, massive capacity, and low latency, 5G could help to support and scale several applications like cloud-connected traffic control, drone delivery, video chatting, and console-quality gaming on the go.

- From global payments and emergency response to distance education and mobile workforce, the benefits and applications of 5G are limitless.

- It has the potential to transform the world of work, the global economy, and people's lives.

Characteristics 5G

- Up to 10Gbps data rate (10 to 100 times speed improvement over 4G and 4.5G networks.)
- Ultra low latency. (1-millisecond latency)
- Massive network capacity.
- More uniform experience to more users
- Increased availability (99.999% availability.)
- 100% coverage.

Who will benefit from 5G?

- 5^{th} generation technology offers a wide range of features, which are beneficial for all group of people including, students, professionals (doctors, engineers, teachers, governing bodies, administrative bodies, etc.) and even for a common man.

- Industries that will benefit the most are:
  - Manufacturing
  - Healthcare
  - Education
  - Government Agencies
  - Transport
  - Crime Detection

Advantages of 5G

- Technology to gather all networks on one platform.

- More effective and efficient.

- Most likely, will provide a huge broadcasting data (in Gigabit), which will support more than 60,000 connections.

- Easily manageable with the previous generations.

- High resolution and bi-directional large bandwidth shaping.

- Possible to provide uniform, uninterrupted, and consistent connectivity across the world

Disadvantages of 5G

- Technology is still under process and research on its viability is going on.

- The speed, this technology is claiming seems difficult to achieve (in future, it might be) because of the incompetent technological support in most parts of the world.

- Many of the old devices would not be competent to 5G, hence, all of them need to be replaced with new one — expensive deal.

- Developing infrastructure needs high cost.

- Security and privacy issue yet to be solved.

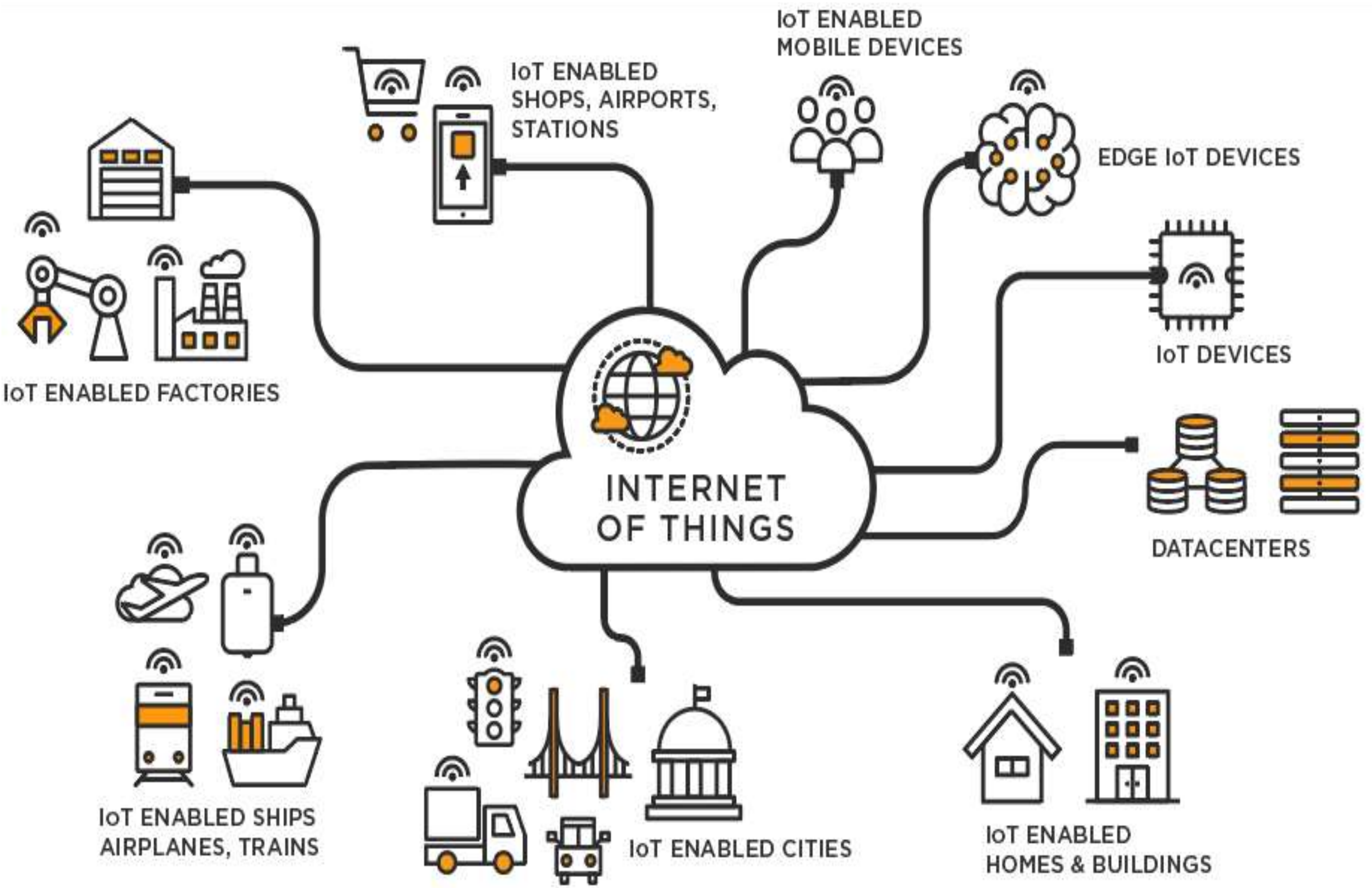
How does 5G works?

- 5G wireless signals are transmitted through large numbers of small cell stations located in places like light poles or building roofs. The use of multiple small cells is necessary because the millimeter wave (mmWave) spectrum-- the band of spectrum between 30 and 300 gigahertz (Ghz) that 5G relies on to generate high speeds -- can only travel over short distances.

- **5G is based on OFDM (Orthogonal frequency-division multiplexing), a method of modulating a digital signal across several different channels to reduce interference.**

- 5G also uses wider bandwidth technologies such as sub-6 GHz and mmWave.

Internet of Things



Internet of Things

- IoT is network of interconnected computing devices which are embedded in everyday objects, enabling them to send and receive data.

- Major Components: Appliances and vehicles, that are **embedded with software, sensors, and connectivity** which enables these objects to connect and exchange data.

- Advancements in medicine, power, gene therapies, agriculture, smart cities, and smart homes are just a very few of the categorical examples where IoT is strongly established.

Internet of Things Characteristics

- Massively scalable and efficient.

- An abundance of physical objects is present that may or may not use IP, so IoT is made possible. Eg: RFID tags, sensors do not use IP address whereas security cameras, smart watches use IP.

- Devices typically consume less power. When not in use, they should be automatically programmed to sleep.

- Intermittent connectivity – IoT devices aren't always connected. In order to save bandwidth and battery consumption, devices will be powered off periodically when not in use.

Internet of Things Applications

- Smart cities
- Smart homes/Home automation
- Healthcare
- Earthquake detection
- Radiation detection/hazardous gas detection
- Smartphone detection
- Water flow monitoring
- Traffic monitoring
- Wearables
- Smart door lock protection system
- Robots and Drones

Advantages of Internet of Things

- Improved efficiency and automation of tasks.
- Increased convenience and accessibility of information.
- Better monitoring and control of devices and systems.
- Greater ability to gather and analyze data.
- Improved decision-making.
- Cost savings.

Disadvantages of Internet of Things

- Security and privacy concerns and potential for hacking.

- Dependence on technology and potential for system failures.

- Limited standardization and interoperability among devices.

- Complexity and increased maintenance requirements.

- High initial investment costs and Limited battery life.

- Concerns about job displacement due to automation.

- Limited regulation and legal framework for IoT, which can lead to confusion and uncertainty.

M2M (Machine-to-Machine)

•M2M technology enables direct communication and data exchange between machines or devices without human intervention, facilitating automated processes and remote monitoring across various industries.

•Common examples include smart home meters, vehicle telemetry services, asset tracking, wearable technologies, automated manufacturing and automated supply chain management (SCM).

•Machines equipped with sensors or other data-gathering tools transmit information over a network to other machines or systems.

M2M (Machine-to-Machine)

- The key difference between M2M (Machine-to-Machine) and IoT (Internet of Things) is that M2M focuses on direct, point-to-point communication between devices, while IoT encompasses a broader network of interconnected devices, people, and services, often involving cloud connectivity and human interaction for enhanced functionality.

- M2M has become a critical component of the Internet of Things (IoT) ecosystem, enabling devices to share information and perform tasks without human intervention.

M2M Advantages

1. **Efficiency and Productivity**:
   1. **Automated Processes**: M2M allows for automated monitoring and management of processes, leading to increased efficiency and productivity.
   2. **Real-time Data**: Devices can communicate in real-time, enabling businesses to make immediate decisions based on accurate data.

2. **Cost Savings**:
   1. **Reduced Labor Costs**: By automating tasks that would otherwise require human intervention, businesses can save on labor costs.
   2. **Predictive Maintenance**: M2M facilitates predictive maintenance, allowing organizations to identify and address issues before they lead to costly breakdowns.

M2M Advantages

3. Enhanced Customer Experience:

1. **Improved Service**: M2M enables businesses to provide better customer service through remote monitoring and troubleshooting.
2. **Personalization**: By analyzing data collected through M2M communication, organizations can personalize products and services to meet customer needs.

4. Scalability:

1. **Expandability**: M2M solutions are scalable, allowing businesses to easily add more devices or integrate with existing systems as their needs evolve.
2. **Integration**: M2M devices can be integrated with various platforms and applications, providing flexibility and interoperability.

M2M Advantages

5. Safety and Security:

1. **Remote Monitoring**: M2M enables remote monitoring of critical infrastructure and assets, enhancing safety and security.
2. **Data Encryption**: Advanced encryption techniques can be employed to secure data transmitted between devices, protecting against unauthorized access and cyber threats.

Disadvantages of M2M

1. **Reliability and Performance**:
    1. **Network Congestion**: As the number of connected devices increases, network congestion and bandwidth limitations may impact performance and reliability.
    2. **Downtime**: System failures, software bugs, and hardware malfunctions can lead to downtime, affecting operations and service availability.

2. **Regulatory and Compliance Issues**:
    1. **Legal Requirements**: Organizations must comply with various regulations and standards governing data privacy, security, and interoperability.
    2. **Data Governance**: Managing data generated by M2M devices requires robust governance policies to ensure compliance with legal and ethical standards.

Disadvantages of M2M

3. **Cost Implications**:
    1. **Initial Investment**: Implementing M2M solutions involves significant upfront costs for hardware, software, and infrastructure.
    2. **Maintenance Costs**: Ongoing maintenance and support can add to the total cost of ownership, especially as the number of devices increases.

4. **Security Concerns**:
    1. **Vulnerabilities**: M2M devices may be susceptible to security vulnerabilities, including malware attacks, data breaches, and unauthorized access.
    2. **Privacy Issues**: Collecting and transmitting data between devices raises privacy concerns related to data ownership, consent, and compliance with regulations.

Disadvantages of M2M

5. **Complexity and Integration Challenges**:
    1. **Compatibility Issues**: Integrating diverse devices and systems can be challenging due to differences in protocols, standards, and technologies.
    2. **Complex Infrastructure**: Building and maintaining a robust M2M infrastructure requires expertise in various domains, including networking, security, and software development.

Differentiate M2M and IoT

M2M	IoT
It is about direct communication between machines	It is about sensor automation and Internet platform
Supports point to point communication	Supports cloud communication
Devices do not necessarily rely on Internet connection	Devices rely on Internet connection or cellular network
Less scalable	More scalable
Doesn’t have human intervention	May have human intervention